Integration

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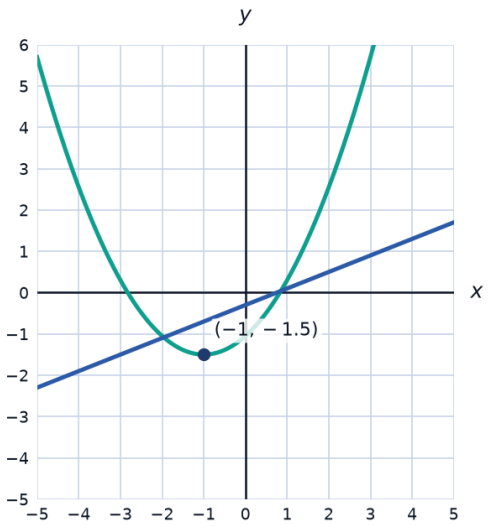
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Integration by parts

Find $\int xe^x\,dx$ using integration by parts.

2 [Foundational] • Target: Integration by parts

Find $\int x^2e^x\,dx$ using integration by parts twice.



3 [Foundational] • Target: Integration using partial fractions

Find $\int \frac{5x+7}{(x+1)(x+2)}\,dx$, first decomposing into partial fractions.

4 [Foundational] • Target: Integration using partial fractions

Find $\int \frac{1}{x^2-4}\,dx$ using partial fractions, factoring $x^2-4=(x-2)(x+2)$.

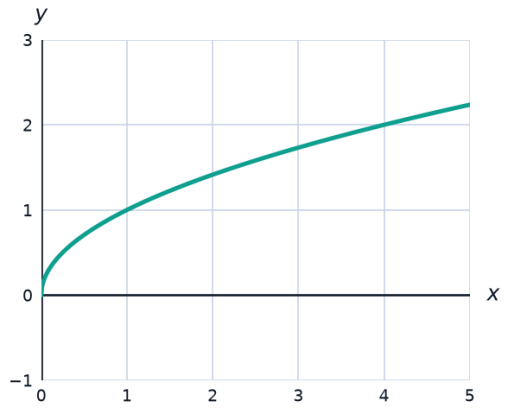
5 [Foundational] • Target: Definite integrals using substitution

Evaluate $\int_0^2 x(x^2+1)^3\,dx$ using the substitution $u=x^2+1$.

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6 [Foundational] • Target: Volumes of revolution: the disk method

The region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis. Find the volume generated, using $V = \pi \int y^2 \, dx$.



7 [Foundational] • Target: Volumes of revolution: the disk method

The region under $y = x^2$ on $[0, 2]$ is revolved about the x-axis. Find the volume.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Volumes of revolution: the disk method

The region under $y = e^x$ on $[0, 1]$ is revolved about the x-axis. Find the volume, using $V = \pi \int_0^1 e^{2x} \, dx$.

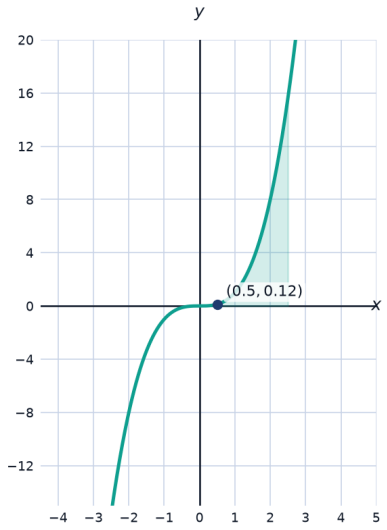
9 [Intermediate] • Target: Volumes of revolution: washer and shell methods

The region between $y = x^2$ and $y = 9$ is revolved about the x-axis. Using the washer method, set up and evaluate the volume integral.

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10 **[Intermediate]** • **Target: Volumes of revolution: washer and shell methods**

The region under $y = x^3$ on $[0, 2]$ is revolved about the y-axis. Use the shell method, $V = 2\pi\int_0^2 x \cdot x^3\,dx$, to find the volume.



11 **[Intermediate]** • **Target: A well-designed word problem: designing a vase**

This question has three parts. A vase is formed by rotating the curve $y = \sqrt{x}$, for $0 \le x \le 9$ (cm), about the y-axis, where x is horizontal distance from the axis and y is height. (a) Express x^2 in terms of y, and evaluate $V = \pi\int_0^3 x^2\,dy$ to find the volume of the vase. (b) A cylindrical base of radius 3 cm and height 2 cm is attached underneath. Find the total volume, to two decimal places (using $\pi \approx 3.14159$). (c) Show that the volume of water filled to height h (ignoring the base) is $V(h) = \frac{\pi h^5}{5}$, and find the height h at which the water volume equals half the vase's volume from part (a), to two decimal places.

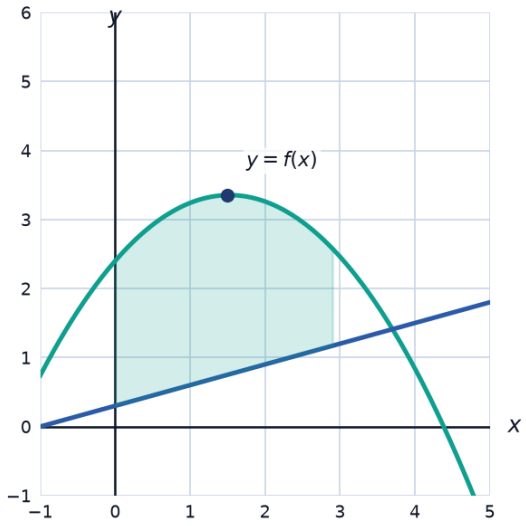
12 **[Intermediate]** • **Target: Integrating trigonometric, exponential and logarithmic functions**

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

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13 [Intermediate] • Target: Definite integrals using substitution

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x \, dx$.



14 [Intermediate] • Target: Volumes of revolution: the disk method

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

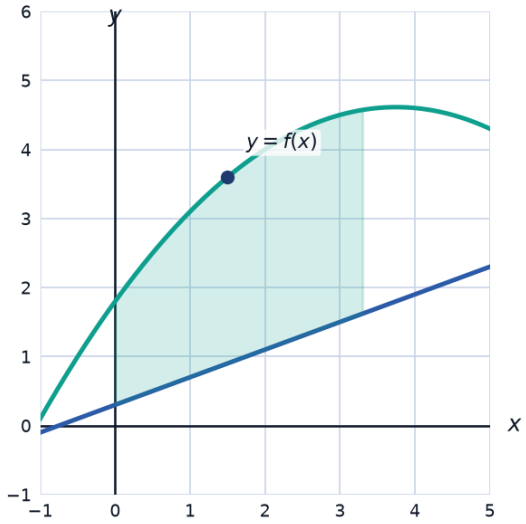
15 [Challenging] • Target: Volumes of revolution: washer and shell methods • Cross-Lesson: 6.01 Sequences & Series

Level 2 · Medium Evaluate the indefinite integral using partial fraction decomposition: $\int \frac{5x - 3}{x^2 - 2x - 3} \, dx$

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16 [Challenging] • Target: A well-designed word problem: designing a vase • Cross-Lesson: 7.01 Vectors

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x \, dx$ for integers $n \geq 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \geq 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x \, dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n \to \infty} I_n$.



17 [Challenging] • Target: Integration by substitution • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Use the substitution $x = 2\sin\theta$ to evaluate the exact definite integral $\int_0^1 \sqrt{4 - x^2} \, dx$.

18 [Challenging] • Target: Integration by parts • Cross-Lesson: 9.01 Complex Numbers

(IB AA/AI HL) Given the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's numerical method with step size $h = 0.1$ to approximate $y(0.2)$. Compare your numerical result with the exact analytical solution.

19 [Challenging] • Target: Integration using partial fractions • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3} \left(x^5 \sqrt{9 - x^2} + \sqrt{9 - x^2} \right) dx$.

20 [Challenging] • Target: Integrating trigonometric, exponential and logarithmic functions • Cross-Lesson: 10.01 Numerical Methods

Use trigonometric substitution to evaluate the indefinite integral $\int \frac{1}{(4 + x^2)^{3/2}} \, dx$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Integration by substitution

$\int \frac{1}{x^2-1}\,dx$ using partial fractions:

a. $\frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| + C$

b. $\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$

c. $-\frac{1}{2}\ln|x^2-1| + C$

d. A and B are equivalent

22 [Foundational] • Target: Integration by parts

$\int_0^{\infty} \frac{1}{1+x^4}\,dx$: evaluate using partial fractions over $\mathbb{R}$:

a. $\pi/2$

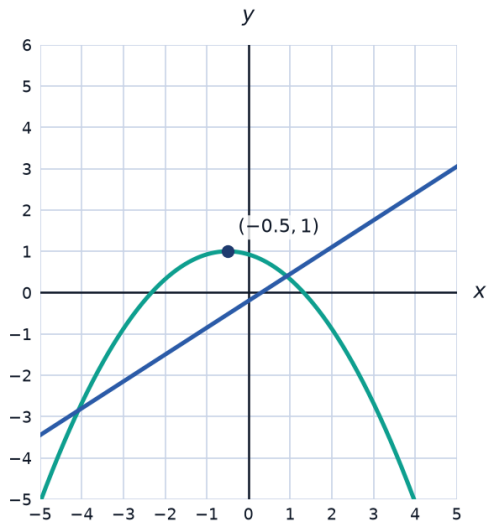
b. $\pi/4$

c. $\pi/(2\sqrt{2})$

d. π

23 [Foundational] • Target: Integration using partial fractions

Shell method for solid of revolution about y-axis: cylindrical shells have radius $=$



a. $f(x)$

b. $b-a$

c. a

d. x

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24 [Foundational] • Target: Integrating trigonometric, exponential and logarithmic functions

Solve $7^x = 100$ for x, expressed using logarithms.

a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$

d. $x = \log_{100}(7)$

25 [Foundational] • Target: Definite integrals using substitution

Volumes: for the same region rotated, which is larger: shell or washer?

a. Shell is always larger

b. They give the same volume

c. Washer is always larger

d. Depends on axis and region

26 [Foundational] • Target: Volumes of revolution: the disk method

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$

27 [Foundational] • Target: Volumes of revolution: washer and shell methods

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.

a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$

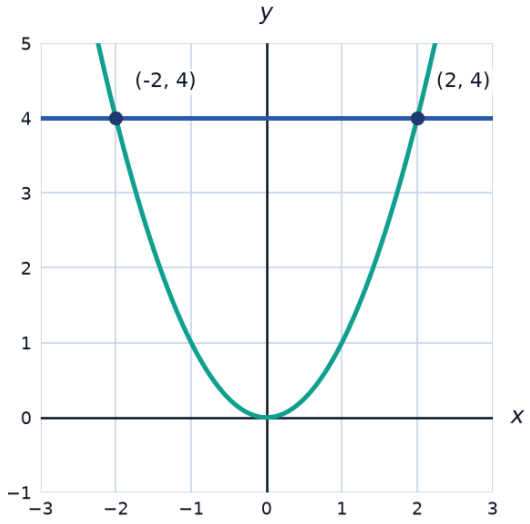
c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

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28 **[Foundational]** • **Target: A well-designed word problem: designing a vase**

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.



a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_{0}^{4}((4 - x^2)^2 - r^2)dx$

c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_{1}^{4}((5 - x^2)^2 - r^2)dx$

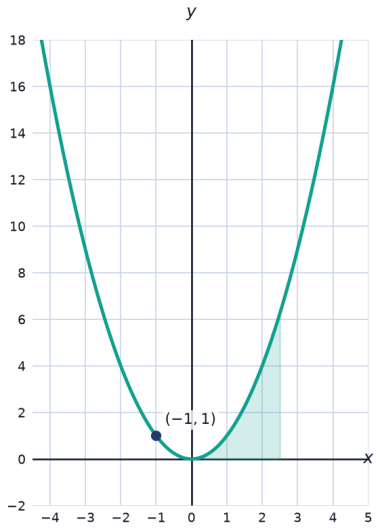
d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Integration by substitution**

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?



a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

30 **[Intermediate]** • **Target: Integration by parts**

Consider the region R bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 0$ and $x = 1$. A solid is formed by rotating R about the y-axis. Determine the volume of this solid using the method of cylindrical shells. Express your answer in terms of π.

a. $\frac{\pi}{2}$

b. $\pi \ln(2)$

c. π

d. $\frac{\pi}{4}$

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31 [Intermediate] • Target: Integration using partial fractions

Consider the region R in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$. A solid is formed by revolving R about the line $x = 5$. Determine the volume of this solid using the method of cylindrical shells. The volume can be expressed in the form $V = k\pi \int_a^b (5 - x)\sqrt{x}\, dx$. Find the value of k.

a. 2

b. 4/3

c. 8/3

d. 16/3

32 [Intermediate] • Target: Integrating trigonometric, exponential and logarithmic functions

Region $x \geq 0$, $y \geq 0$, $x + y \leq 1$: volume when rotated about y-axis by shell method:

a. $\frac{\pi}{3}$

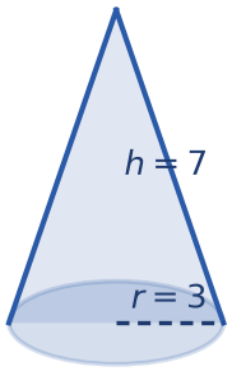
b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$

d. $\frac{\pi}{6}$

33 [Intermediate] • Target: Definite integrals using substitution

Washer method: volume when $R(x)$ is outer and $r(x)$ inner radius:



a. $\pi \int (R - r)^2 dx$

b. $\pi \int (R^2 - r^2) dx$

c. $2\pi \int (R - r) dx$

d. $\pi \int R^2 dx - \pi \int r^2 dx$ (same)

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34 [Intermediate] • Target: Volumes of revolution: the disk method

Repeated linear factor in partial fractions: $\frac{1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. Decompose $\frac{x}{(x-1)^2}$:

a. $\frac{1}{x-1} + \frac{1}{(x-1)^2}$

b. $\frac{x}{(x-1)^2}$

c. $\frac{1}{x-1}$

d. $\frac{2}{(x-1)^2}$

35 [Intermediate] • Target: Volumes of revolution: washer and shell methods

Shell method: volume when shell radius $= x$, height $= f(x)$:

a. $2\pi\int_a^b xf(x)dx$

b. $\int_a^b f(x)dx$

c. $\pi\int_a^b x^2 f(x)dx$

d. $\pi\int_a^b f(x)^2 dx$

36 [Intermediate] • Target: A well-designed word problem: designing a vase

Volume of cone: radius r, height h, using disc method:

a. $\frac{2}{3}\pi r^2 h$

b. $\frac{1}{2}\pi r^2 h$

c. $\pi r^2 h$

d. $\frac{1}{3}\pi r^2 h$

37 [Intermediate] • Target: Integration by substitution

Volume of solid of revolution when region $0 \le y \le e^{-x^2}$, $0 \le x \le 1$ is rotated about y-axis (shell):

a. $\pi(1 - e^{-1})$

b. $\pi(e-1)/e$

c. $\pi/2$

d. $2\pi\int_0^1 xe^{-x^2}dx$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Integration by parts • Cross-Lesson: 9.01 Complex Numbers

Find $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$ using partial fractions:

a. 1

b. 1/2

c. 2/3

d. 3/4

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39 [Challenging] • Target: Integration using partial fractions • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Volume of revolution using shells: $y = \cos(x^2)$, $x \in [0, \sqrt{\pi/2}]$, rotated about y-axis:

a. 2π

b. $\pi/2$

c. π

d. $\sqrt{\pi}$

40 [Challenging] • Target: Integrating trigonometric, exponential and logarithmic functions • Cross-Lesson: 10.01 Numerical Methods

A population of 500 grows at 5% per year, modeled by $P(t) = 500(1 + 0.05)^t$. Approximately how many years until the population reaches $1.5\times$ its initial size (i.e. $P = 1.5 \cdot 500$)?

a. 8.3 years

b. 10.8 years

c. 6.3 years

d. 10.3 years

41 [Challenging] • Target: Definite integrals using substitution • Cross-Lesson: 11.01 Differential Equations

Surface area of sphere radius R using surface area of revolution:

a. $4\pi R^2$

b. $3\pi R^2$

c. $2\pi R^2$

d. πR^3

42 [Challenging] • Target: Volumes of revolution: the disk method • Cross-Lesson: 12.01 Poisson Distribution

Integration by parts: $\int xe^x\,dx =$

a. $e^x(x-1) + C$

b. $e^x(x+1) + C$

c. $xe^x + C$

d. $\frac{x^2e^x}{2} + C$

43 [Challenging] • Target: Volumes of revolution: washer and shell methods • Cross-Lesson: 13.01 Continuous Random Variables

Write $5^4 = 625$ in logarithmic form.

a. $\log_{625}(5) = 4$

b. $\log_5(4) = 625$

c. $\log_4(5) = 625$

d. $\log_5(625) = 4$

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44 [Challenging] • Target: A well-designed word problem: designing a vase • Cross-Lesson: 14.01 Sampling & Estimation

Partial fractions: $\frac{2x+3}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2}$. Find A and B:

a. $A = 2, B = 1$

b. $A = 1, B = 2$

c. $A = 2, B = 3$

d. $A = 3, B = 2$

45 [Challenging] • Target: Integration by substitution • Cross-Lesson: 15.01 Hypothesis Testing

The method of Lagrange multipliers finds constrained extrema using multiplier λ. The critical value of λ itself:

a. is the rate of change of the optimal value w.r.t. the constraint constant

b. gives the extremum value

c. is always zero

d. equals $|ablaf|$

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